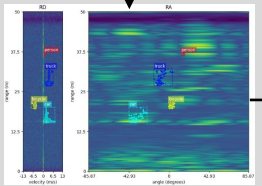
1) Radar Target Annotation



Stereo Project



3D BBox (x,y,z,w,h,d)
Object Class

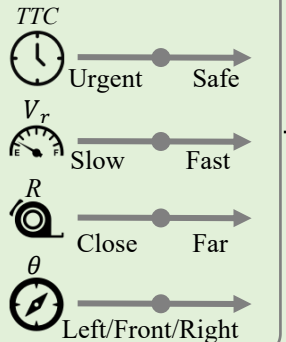
2) Semantic Quantization of Radar Attributes

(a) Physical Measurement Calculation

$$(i, j, k) = f_{\text{center}}(\text{BBox})$$

$$\begin{cases} R = R_{max} - i \cdot \Delta_r \\ \theta = (j - N_A/2) \cdot \Delta_a \\ V_r = (k - N_D/2) \cdot \Delta_v \\ TTC = \frac{R}{V_r} \end{cases}$$

(b) Semantic Quantization



3) Referring Expression Generation


```
{
 "dataID": "part_1/000029",
 "Class": "car",
 "range": "close distance",
 "azimuth": "front left",
 "speed": "moderate speed",
 "action": "approaching",
 "TTC_zone": "safe"
}
```


GPT-4

Referring Expression

"A car is driving approaching at a moderate speed from the front left."

"Pinpoint the car on the front left driving at a safe moderate speed."